

PCD Assignment 01 Report Analysis

Image 1: image1_colorful.jpg

Colorful Image: Downsampling + Upsampling Reconstructions



This is an image of a colorful garden containing many saturated colors, small flowers, leaves, and dense textures. These characteristics make it useful for observing color preservation, texture loss, and smoothing after sampling.

- Experimental Setup

The original image has a cropped-even resolution of 2048×1260 pixels. It is reduced by a factor of 2 using Max, Average, and Median downsampling. Each reduced image is then restored to the original dimensions using Nearest Neighbor, Bilinear, and Bicubic interpolation.

- Result and Analysis

- The best quantitative reconstruction was Median + Bicubic, with $MSE = 25.62$ and $PSNR = 34.05$ dB.
- The weakest combination was Max + Nearest Neighbor, with $MSE = 147.75$ and $PSNR = 26.44$ dB.

Average and median downsampling preserve the overall color distribution better than max downsampling. The large number of colorful small objects means that reduction inevitably removes fine flower details, but bicubic interpolation produces the smoothest transitions when the image is

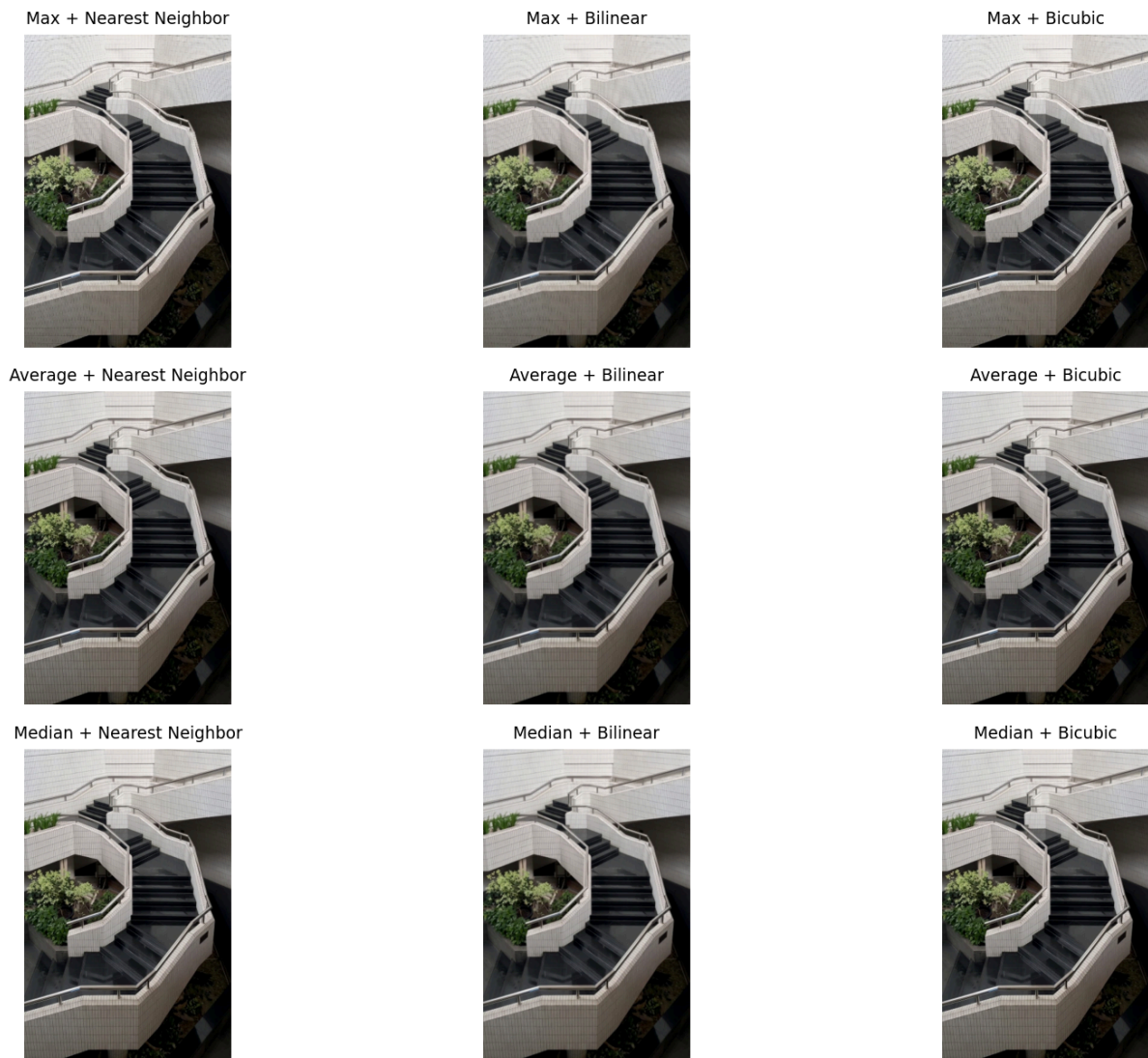
reconstructed. Max downsampling tends to emphasize the strongest channel values in each block, which can create noticeable color and brightness changes.

- Conclusion

This image demonstrates that sampling quality depends strongly on image content. Downsampling permanently removes spatial information, while upsampling can only estimate the missing pixels. Therefore, the choice of downsampling and interpolation method should consider whether the image is dominated by color variation, sharp edges, or fine repetitive textures.

Image 2: image2_staircase.jpeg

Staircase Image: Downsampling + Upsampling Reconstructions



The staircase contains strong straight edges, repeated tile lines, dark steps, and high-contrast boundaries. It is especially useful for observing aliasing, edge sharpness, blockiness, and interpolation artifacts

- Experimental Setup

The original image has a cropped-even resolution of 688×1044 pixels. It is reduced by a factor of 2 using Max, Average, and Median downsampling. Each reduced image is then restored to the original dimensions using Nearest Neighbor, Bilinear, and Bicubic interpolation.

- Result and Analysis
 - The best quantitative reconstruction was Median + Bicubic, with MSE = 89.55 and PSNR = 28.61 dB.
 - The weakest combination was Max + Nearest Neighbor, with MSE = 284.40 and PSNR = 23.59 dB.

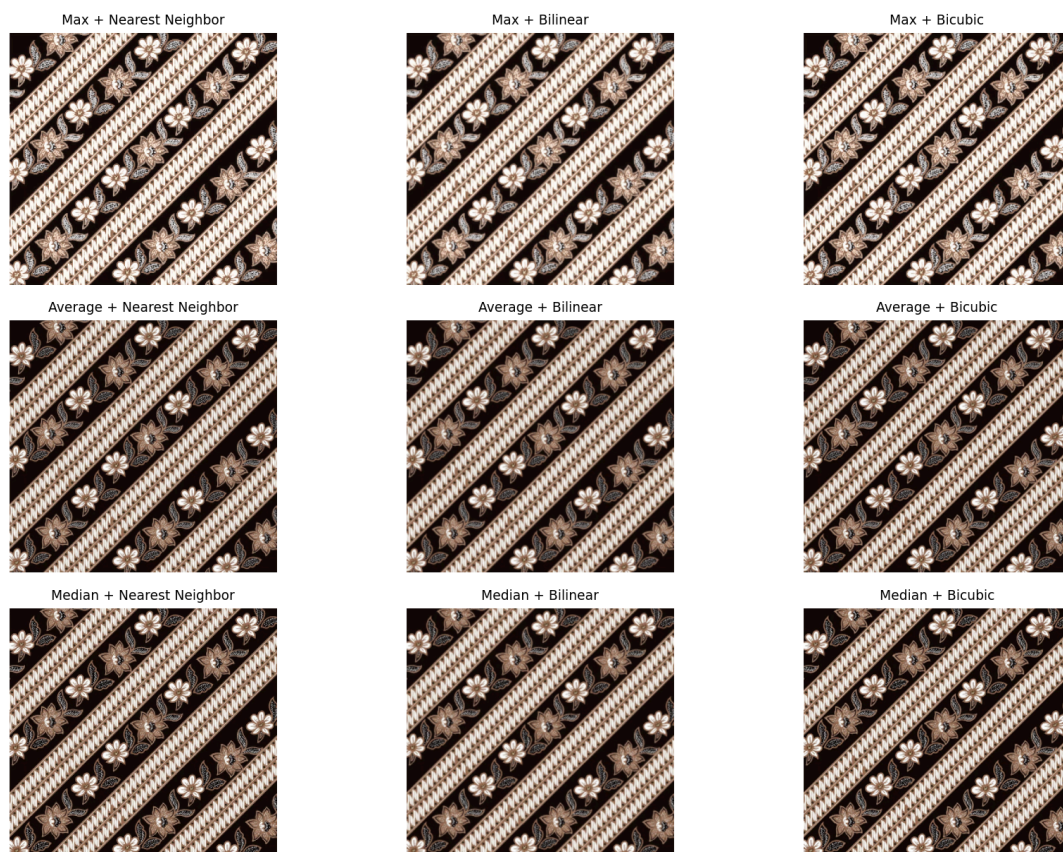
The staircase is highly sensitive to edge distortion because its visual structure is dominated by straight lines and sharp boundaries. Max downsampling produces larger reconstruction errors, while average and median retain the overall geometry more consistently. Bicubic gives the highest PSNR in the experiment and visually softens transitions, although excessive smoothing can slightly reduce the crispness of tile lines and stair edges.

- Conclusion

This image demonstrates that sampling quality depends strongly on image content. Downsampling permanently removes spatial information, while upsampling can only estimate the missing pixels. Therefore, the choice of downsampling and interpolation method should consider whether the image is dominated by color variation, sharp edges, or fine repetitive textures.

Image 3: image3_batik.jpg

Batik Image: Downsampling + Upsampling Reconstructions



The batik image contains repetitive diagonal motifs, thin decorative lines, flowers, and relatively uniform background regions. It is useful for observing the preservation of fine repetitive patterns and color boundaries.

- Experimental Setup

The original image has a cropped-even resolution of 626×590 pixels. It is reduced by a factor of 2 using Max, Average, and Median downsampling. Each reduced image is then restored to the original dimensions using Nearest Neighbor, Bilinear, and Bicubic interpolation.

- Result and Analysis

- The best quantitative reconstruction was Average + Bicubic, with $MSE = 725.48$ and $PSNR = 19.52$ dB.
- The weakest combination was Max + Nearest Neighbor, with $MSE = 2272.59$ and $PSNR = 14.57$ dB.

The batik pattern contains repeated fine lines and diagonal motifs, so downsampling causes loss of small decorative details and local color transitions. Average downsampling achieves the highest PSNR for this image, indicating that averaging provides a good balance between preserving the overall pattern and reducing local variation. Bicubic reconstruction performs best among the three upsampling methods, but some fine motifs cannot be recovered because information lost during downsampling cannot be recreated exactly.

- Conclusion

This image demonstrates that sampling quality depends strongly on image content. Downsampling permanently removes spatial information, while upsampling can only estimate the missing pixels. Therefore, the choice of downsampling and interpolation method should consider whether the image is dominated by color variation, sharp edges, or fine repetitive textures.